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M150
Problem set 2

1a) standard SIR:

$$\frac{dS}{dt} = -BSI \qquad \frac{dI}{dt} = BSI - \gamma I \qquad \frac{dR}{dt} = \gamma I$$

rate of sus. — rate of infected — rate of recovered

B = avg rate at which infecties are affecting susceptibles
p = avg. recovery rate when infecties are cared for by those who are recovered
γ = recovery rate of infecties when received no care

$p \cdot I \cdot R$ => interaction between infecties and recovered individuals multiplied by p

$$\boxed{I} \xrightarrow{pIR} \boxed{R}$$

so modified equations:

$$\frac{dS}{dt} = -BSI \qquad \frac{dI}{dt} = BSI - \gamma I - pIR \qquad \frac{dR}{dt} = \gamma I + pIR$$

1b)

$$N = S + I + R$$

↳ constant or change in time?

$$\frac{dN}{dt} = \frac{dS}{dt} + \frac{dI}{dt} + \frac{dR}{dt}$$

$$\frac{dN}{dt} = -\cancel{BSI} + \cancel{BSI} - \cancel{\gamma I} - \cancel{pIR} + \cancel{\gamma I} + \cancel{pIR}$$

$$\frac{dN}{dt} = 0$$

↳ change in N w/respect to time is 0

↳ in other words $\Rightarrow$ $\boxed{N = \text{constant}}$

Since N = constant

$$N = S + I + R$$

↳ $S = N - I - R$

↳ since S can be computed algebraically from I and R

↳ $\boxed{\text{3 system equation} \xrightarrow{\text{reduction}} \text{solving 2 ODEs}}$

↳ $\frac{dI}{dt}$ and $\frac{dR}{dt}$

$$\frac{dI}{dt} = B(N - I - R)I - \gamma I - pRI$$

$$\frac{dR}{dt} = \gamma I + pRI$$

1c)

where nothing is changing

fixed points → set reduced equations = 0

→ $\frac{dI}{dt} = 0$ and $\frac{dR}{dt} = 0$

$$\frac{dI}{dt} = B(N-I-R)I - \gamma I - pIR = 0$$

$$I(B(N-I-R) - \gamma - PR) = 0$$

$I = 0$ or $BS - \gamma - PR = 0$

$PR = BS - \gamma$

$R = \frac{BS - \gamma}{P}$

→ this branch does not matter since $I=0$ and $\frac{dI}{dt} = 0$ satisfied for any value of R when $I = 0$

$$\frac{dR}{dt} = \gamma I + pIR = 0$$

$$I(\gamma + PR) = 0$$

$I = 0$ or $\gamma + PR = 0$

only valid case

$PR = -\gamma$

$R = \frac{-\gamma}{P}$ → invalid

$$\boxed{(I^*, R^*) = (0, R^*) \text{ for any } R^* \in (0, N)}$$

1d)

$$J = \begin{pmatrix} \frac{\partial}{\partial I}\frac{dI}{dt} & \frac{\partial}{\partial R}\frac{dI}{dt} \\ \frac{\partial}{\partial I}\frac{dR}{dt} & \frac{\partial}{\partial R}\frac{dR}{dt} \end{pmatrix}$$

$$\frac{dI}{dt} = B(N-I-R)I - \gamma I - pRI \qquad \frac{dR}{dt} = \gamma I + pRI$$

$$\frac{dI}{dt} = BNI - BI^2 - BRI - \gamma I - pRI$$

$$\frac{\partial}{\partial I}\frac{dI}{dt} = BN - 2BI - BR - \gamma - pR$$
$$= B(N-2I-R) - \gamma - pR$$

$$\frac{\partial}{\partial R}\frac{dI}{dt} = -BI - pI$$

$$\frac{\partial}{\partial I}\frac{dR}{dt} = \gamma + pR$$

$$\frac{\partial}{\partial R}\frac{dR}{dt} = pI$$

$$\boxed{J = \begin{pmatrix} B(N-2I-R) - \gamma - pR & -BI - pI \\ \gamma + pR & pI \end{pmatrix}}$$

1e)

newly introduced disease means:

$$\boxed{I = 0, \quad R = 0, \quad S = N}$$

→ no infection or recovery has happened yet

# of susceptible = population N

letting $I = 0$, $R = 0$ into J

$$\boxed{J\Big|_{\substack{I=0,\\ R=0}} = \begin{pmatrix} \beta N - \gamma & 0 \\ \gamma & 0 \end{pmatrix}}$$

1f)

$$J\Big|_{\substack{I=0,\\ R=0}} = \begin{pmatrix} BN-\gamma & 0 \\ \gamma & 0 \end{pmatrix}$$

$$\det(J-\lambda I)=0$$

$$\hookrightarrow \begin{pmatrix} BN-\gamma & 0 \\ \gamma & 0 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} = \begin{pmatrix} BN-\gamma-\lambda & 0 \\ \gamma & -\lambda \end{pmatrix}$$

$$\det\begin{pmatrix} BN-\gamma-\lambda & 0 \\ \gamma & -\lambda \end{pmatrix} = 0$$

$$\hookrightarrow (BN-\gamma-\lambda)(-\lambda) - (\lambda)(0) = 0$$

$$-BN\lambda + \gamma\lambda + \lambda^2 = 0$$

$$\lambda^2 - BN\lambda + \gamma\lambda = 0$$

$$\lambda(\lambda - BN + \gamma) = 0$$

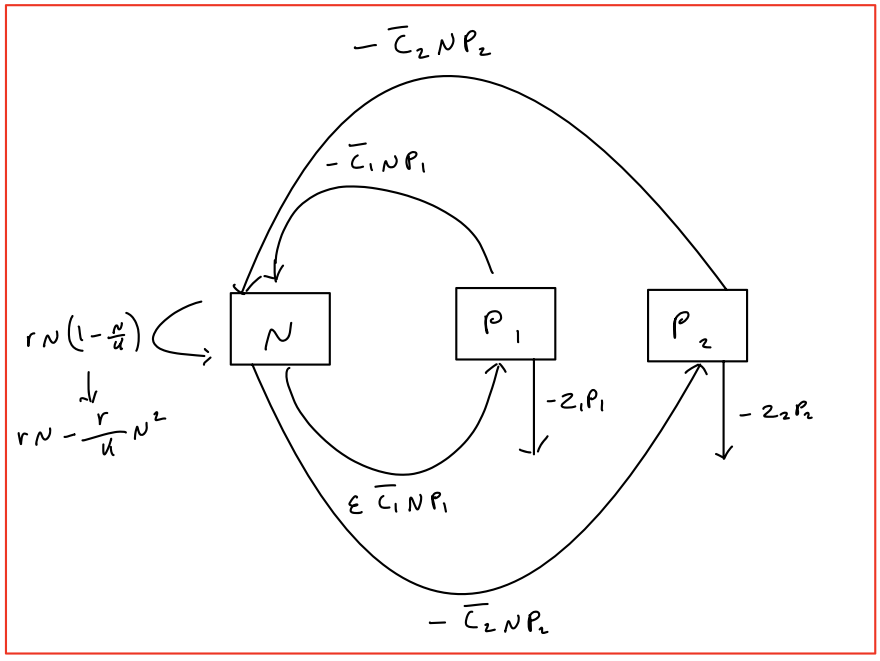
$$\hookrightarrow \lambda - BN + \gamma = 0$$

$$\lambda = BN - \gamma$$

$$\boxed{\begin{aligned} \lambda_1 &= 0 \\ \lambda_2 &= BN-\gamma \end{aligned}}$$

2a)

network diagram



Interaction Matrix

$$M_I \vec{X} = \begin{bmatrix} -\frac{r}{K} & -\bar{C}_1 & -\bar{C}_2 \\ \varepsilon \bar{C}_1 & 0 & 0 \\ \varepsilon \bar{C}_2 & 0 & 0 \end{bmatrix} \begin{bmatrix} N \\ P_1 \\ P_2 \end{bmatrix} =$$

Trace = sum of diagonal entries

$$T_{M_I} = \left(-\frac{r}{K} + 0 + 0\right) = -\frac{r}{K}$$

$$\boxed{\text{Trace} = -\frac{r}{K} < 0 \text{ (negative)}}$$

$$\text{Det}(M_I) =$$

$$\left(-\bar{C}_1\left(\varepsilon \bar{C}_1 \cdot 0\right) - \left(\varepsilon \bar{C}_2 \cdot 0\right)\right) - \left(-\frac{r}{K}\left(0 \cdot 0\right) - \left(0 \cdot 0\right)\right) - \left(-\bar{C}_2\left(\varepsilon C_1 \cdot 0\right) - \left(\varepsilon \bar{C}_2 \cdot 0\right)\right)$$

$$= 0$$

$$\boxed{\text{Det}(M_I) = 0}$$

2b)

fixed points → change equations = 0

$$\frac{dP_1}{dt} = 0$$

$$\varepsilon \bar{C}_1 N P_1 - z_1 P_1 = 0$$

$$P_1(\varepsilon \bar{C}_1 N - z_1) = 0$$

$$P_1 = 0 \quad \text{or} \quad N = \frac{z_1}{\varepsilon \bar{C}_1}$$

$$\varepsilon \bar{C}_1 N - z_1 = 0$$

$$\varepsilon \bar{C}_1 N = z_1$$

$$N = \frac{z_1}{\varepsilon \bar{C}_1}$$

$$\frac{dP_2}{dt} = 0$$

$$\varepsilon \bar{C}_2 N P_2 - z_2 P_2 = 0$$

$$P_2(\varepsilon \bar{C}_2 N - z_2) = 0$$

$$P_2 = 0 \quad \text{or} \quad N = \frac{z_2}{\varepsilon \bar{C}_2}$$

$$\frac{dN}{dt} = 0$$

$$rN\left(1 - \frac{N}{K}\right) - \bar{C}_1 N P_1 - \bar{C}_2 N P_2 = 0$$

$$N\left(r\left(1 - \frac{N}{K}\right) - \bar{C}_1 P_1 - \bar{C}_2 P_2\right) = 0$$

$$r\left(1 - \frac{N}{K}\right) - \bar{C}_1 P_1 - \bar{C}_2 P_2 = 0$$

$$r\left(1 - \frac{N}{K}\right) = \bar{C}_1 P_1 + \bar{C}_2 P_2$$

$$N = 0 \quad \text{or} \quad r\left(1 - \frac{N}{K}\right) = \bar{C}_1 P_1 + \bar{C}_2 P_2$$

$$P_2^* = 0$$

$$r\left(1 - \frac{\frac{z_1}{\varepsilon C_1}}{\frac{K}{1}}\right) = \bar{C}_1 P_1 + \bar{C}_2 (0)$$

$$r - r\left(\frac{z_1}{\varepsilon C_1 K}\right) = \bar{C}_1 P_1$$

$$r\left(1 - \frac{z_1}{\varepsilon C_1 K}\right) = \bar{C}_1 P_1$$

$$\frac{r}{\bar{C}_1}\left(1 - \frac{z_1}{\varepsilon C_1 K}\right) = P_1$$

There is no fixed point that allows 3 species to exist at the same time. One species must always go extinct.

Fixed Point	N^*	P_1^*	P_2^*	
FP1	0	0	0	extinction
FP2	K	0	0	carrying capacity
FP3	$\frac{z_1}{\varepsilon C_1}$	$\frac{r}{\bar{C}_1}\left(1 - \frac{z_1}{\varepsilon C_1 K}\right)$	0	prey + P_1
FP4	$\frac{z_2}{\varepsilon C_2}$	0	$\frac{r}{\bar{C}_2}\left(1 - \frac{z_2}{\varepsilon C_2 K}\right)$	prey + P_2

2c)

$$P_2 > 0 \rightarrow FP4$$

$$FP4 = (N^*, P_1^*, P_2^*) = \left(\frac{z_2}{\varepsilon c_2}, 0, \frac{r}{\bar{c}_2}\left(1 - \frac{z_2}{\varepsilon c_2 u}\right)\right)$$

since $N^* = \frac{z_2}{\varepsilon c_2}$

$$(N^*, P_1^*, P_2^*) = \left(\frac{z_2}{\varepsilon c_2}, 0, \frac{r}{\bar{c}_2}\left(1 - \frac{N^*}{u}\right)\right)$$

$$J = \begin{bmatrix} r - \frac{2rN}{u} - \bar{c}_1 P_1 - c_2 P_2 & -\bar{c}_1 N & -\bar{c}_2 N \\ \varepsilon \bar{c}_1 P_1 & \varepsilon \bar{c}_1 N - z_1 & 0 \\ \varepsilon \bar{c}_2 P_2 & 0 & \varepsilon \bar{c}_2 N - z_2 \end{bmatrix}$$

since: $P_2^* = \frac{r}{\bar{c}_2}\left(1 - \frac{N^*}{u}\right)$

$$\bar{c}_2 P_2^* = r\left(1 - \frac{N^*}{u}\right)$$

$$(1,1) = r - \frac{2rN^*}{u} - \bar{c}_2 P_2^* = r - \frac{2rN^*}{u} - r\left(1 - \frac{N^*}{u}\right)$$

$$= \cancel{r} - \frac{2rN^*}{u} - \cancel{r} + \frac{rN^*}{u} = \frac{-rN^*}{u}$$

$$(3,3) = \varepsilon \bar{c}_2 N - z_2 = \varepsilon \bar{c}_2 N^* - z_2$$

$$= \cancel{\varepsilon}\cancel{\bar{c}_2}\left(\frac{z_2}{\cancel{\varepsilon}\cancel{\bar{c}_2}}\right) - z_2 = z_2 - z_2 = 0$$

$$\boxed{J(N^*, P_1^*, P_2^*) = \begin{bmatrix} \frac{-rN^*}{u} & -\bar{c}_1 N^* & -\bar{c}_2 N^* \\ 0 & \varepsilon \bar{c}_1 N^* - z_1 & 0 \\ \varepsilon \bar{c}_2 P_2^* & 0 & 0 \end{bmatrix}}$$

where $N^* = \frac{z_2}{\varepsilon c_2}$ and $P_2^* = \frac{r}{\bar{c}_2}\left(1 - \frac{N^*}{u}\right)$

2d)

$$Tr(J) = \frac{-rN^*}{K} + (\varepsilon \bar{c}_1 N^* - z_1) + 0$$

where $N^* = \frac{z_2}{\varepsilon c_2}$ so . . .

$$\frac{-r\left(\frac{z_2}{\varepsilon c_2}\right)}{K} + \cancel{\varepsilon}\bar{c}_1\left(\frac{z_2}{\cancel{\varepsilon} c_2}\right) - z_1$$

$$Tr(J) = \frac{-r z_2}{\varepsilon c_2 K} + \frac{\bar{c}_1 z_2}{c_2} - z_1$$

- Trace sign depends on parameter values
- r, N^*, $\frac{1}{K}$ enter the stabilizing term of $\frac{rN^*}{K}$ lead to larger prey growth rate, higher equilibrium prey density, and stricter carrying capacity: all of which push the trace more negative.
- $\bar{c}_1$ pulls the trace upward which is destabilizing. A more efficient P_1 consumer can extract energy from prey faster.
- $\bar{z}_1$ pulls the trace downward which is stabilizing since higher death rate on P_1 prevents it from invading.
- z_2, $\bar{c}_2$ enter through N^* and set the prey level at the fixed point N^*. Small $z_2/\bar{c}_2$ keeps N^* low if predator P_2 needs little prey to break even. This suppresses $\varepsilon \bar{c}_1 N^*$, pushing the trace negative. In other words, P_2 holding the prey down protects itself from an invasion by P_1.

2e) P_2 => mutualist of prey

$$\frac{dN}{dt} = rN\left(1-\frac{N}{K}\right) - \bar{C}_1 N P_1 \boxed{+ \bar{B}_2 N P_2}$$

sign of prey - P_2 interaction flips in dN/dt where $\bar{B}_2$ (benefit coefficient) is greater than zero.

$$\frac{dP_1}{dt} = \varepsilon \bar{C}_1 N P_1 - z_1 P_1$$

$$\frac{dP_2}{dt} = \varepsilon \bar{C}_2 N P_2 - z_2 P_2$$

3) $\boxed{\text{NO}}$

- Interaction matrix captures effect of species i on species j as a pairwise linear reaction. In other words, interaction matrix built on 2-body interactions where two species are involved at exact same time.

  Ex. => $\alpha E_1 P_1$ or $\beta E_2 P_1$

  ↳ each involve 2 species

- However, the reaction needs E_1, E_2, and P_1 to all three be present at the same time.

  Ex. => $\alpha E_1 E_2 P_1$

  ↳ cubic, three-body interaction term and cannot be decomposed into any sum of pairwise interactions. That being said, no interaction matrix ever can capture it.